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Molecular evidence and diversity of the spotted-fever group *Rickettsia* spp. in small mammals from natural, suburban and urban areas of Eastern Slovakia 2

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ABSTRACT 12

Rickettsiae of the spotted fever group are considered as emerging pathogens; ticks, fleas and mites are known to be their vectors. However, the prevalence and species diversity of rickettsiae in small mammals and the role of these hosts in the circulation of bacteria are much less studied. During 2014–2016, a total of 250 small mammals (*Apodemus agrarius*, *Apodemus flavicollis*, *Apodemus uralensis*, *Myodes glareolus*, *Crocidura leucodon*, *Crocidura suaveolens*, *Micromys minutus*, *Microtus arvalis*, *Microtus subterraneus*, *Neomys fodiens* and *Sorex minutus*) were captured in natural, suburban and urban habitats of eastern Slovakia. Ear biopsies of 245 individuals were examined for the presence of SFG rickettsiae by molecular methods. The overall prevalence of *Rickettsia* spp. in small mammals was 11%. The predominant species, *Rickettsia helvetica* was confirmed in the striped field mouse (*A. agrarius*), the yellow-necked mouse (*A. flavicollis*), the bank vole (*M. glareolus*) and the common vole (*M. arvalis*) in natural and suburban habitats, followed by *Rickettsia* sp. closely related to *R. felis* identified in *A. flavicollis* captured in a suburban habitat. Finally, *R. slovaca* was found in only one yellow-necked mouse (*A. flavicollis*) captured in a natural habitat, with the sympatric occurrence of *Dermacentor marginatus* and *Dermacentor reticulatus* ticks. We assume the presence of *R. slovaca* especially in sites with the occurrence of *Dermacentor* spp. All small mammals captured in the urban habitat tested were negative for the presence of rickettsiae. This study brings the first molecular evidence of *R. slovaca* in a rodent captured in Slovakia. *Rickettsia* sp. closely related to *R. felis* was first time detected in *A. flavicollis* in suburban site of Slovakia. 13

The highest species diversity of rickettsiae was observed in *A. flavicollis*, and the highest prevalence of bacteria was recorded in *M. glareolus*. The highest occurrence of rickettsiae-positive small mammals was recorded during the spring and autumn months, May, June and September, respectively. This may be related with the seasonal activity of the tick vectors. This study confirms the long-term persistence of *Rickettsia* spp. in small mammals in natural and suburban habitats of Slovakia. 14

Some rodent species that have a wider ecological valency may contribute to the maintenance, circulation and dissemination of rickettsiae within and out the natural foci more significantly than those species that have narrower relation to the certain type of habitat. 15

1. Introduction 16

The genus *Rickettsia* (Rickettsiaceae, Rickettsiales) includes Gram-negative obligate, intracellular, aerobic α-Proteobacteria invading hosts' eukaryotic cells. Species of the genus *Rickettsia* are divided into two groups: the spotted fever group (SFG) and the typhus group (TG) (Raoult et al., 2005). The main vectors of rickettsiae are blood-sucking arthropods, mainly ticks, but also fleas, mites and human body lice. The 17 highest occurrence of rickettsiae in ticks was detected in early spring and autumn (Řeháček et al., 1972). Řeháček et al. (1972) suggested that rickettsiae are able to hibernate in ticks. The most important vectors and probably also the main reservoirs of rickettsiae in Southern, Western and Central Europe are *Dermacentor marginatus* and *Dermacentor reticulatus* ticks (Řeháček et al., 1972, 1977; Raoult et al., 2002). 18

Small rodents, hosting a wide spectrum of ectoparasites and arthropod vectors, are the most significant reservoirs of rickettsiae, many 19

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of which are responsible for serious zoonotic diseases (rickettsioses) of humans and animals (Řeháček et al., 1992; Burri et al., 2014; Han et al., 2015; Imhoff et al., 2015). Rickettsioses belong among the most frequent vector-borne infections in Europe (Fournier et al., 2012).

In Slovakia, six rickettsial species/strains of the SFG have thus far been identified by serological examination and/or different molecular methods. Among the identified species are *Rickettsia helvetica, Rickettsia slovaca*, *Rickettsia raoultii*, *Rickettsia monacensis* strain IRS3 and IRS4, *Rickettsia conorii* and *Rickettsia africae* (Brezina et al., 1969; Řehácek et al., 1975; Boldiš et al., 2008; Špitalská et al., 2012, 2014; Sekeyová et al., 2013). Recently, *Rickettsia* sp., closely related to *Rickettsia felis*, was detected in Slovakia (Špitalská et al., 2015). These species of rickettsiae are usually transmitted by the ticks, but *R. africae* and *R. felis* have also been identified in fleas (Sekeyová et al., 2013; Špitalská et al., 2015).

R. slovaca was isolated for the first time in 1968 from *D. marginatus* ticks in Central Slovakia (Brezina et al., 1969). After the first evidence of this agent in vector ticks, several patients with clinical signs similar to those associated with *R. slovaca* infection, confirmed in a French patient in 2003, have been reported (Parola et al., 2009; Cascio et al., 2013). Based on the most significant clinical symptoms in affected patients, such as eschar and cervical lymphadenopathies, several names have been proposed for diseases resulting from human infections, e.g. tick-borne lymphadenopathy (TIBOLA), *Dermacentor*-borne necrosis-erythrema-lymphadenopathy (DEBONEL) and/or scalp eschar and neck lymphadenopathy (SENLAT) (Lakos, 1997, 1999; Oteo et al., 2003; Angelakis et al., 2010). The peak incidence of human rickettsioses usually occurs from March to May and during September and November, except the dry summer season, and this is clearly connected with the seasonal activity of *Dermacentor* ticks (Estrada-Peña et al., 2004; Parola et al., 2009).

In Slovakia, the *Ixodes ricinus* tick is considered the main vector and natural reservoir of *R. helvetica* (Špitalská et al., 2008, 2014; Sekeyová et al., 2013).

Rickettsia felis is primarily associated with an arthropod reservoir and vector, the cat flea *Ctenocephalides felis*. It causes infection in humans, known as flea-borne spotted fever or cat-flea typhus (Abdad et al., 2011). In Slovakia, *Rickettsia* sp. closely related to *R. felis* was detected for the first time in a *Ctenophthalmus solutus* flea removed from a striped field mouse (*A. agrarius*) (Špitalská et al., 2015).

In Slovakia, research on rickettsiae has a long tradition (Sekeyová et al., 2013), but the role of rodents in their circulation has been much less studied by molecular methods so far. Recently, only a few studies referring to the presence of *R. helvetica* and *R. monacensis* in *Apodemus flavicollis* and/or *M. arvalis* (Minichová et al., 2017), and *Rickettsia* spp. in *M. arvalis* (Smetanová et al., 2006) have been published.

In the works of Jablonskaja (1978), *R. slovaca* was found in three dominant rodent species, *A. agrarius*, *A. flavicollis* and *M. arvalis*, in Eastern Slovakia; however, these results were only based on serological analysis and cross-immunity studies, without further molecular evidence. Řeháček et al. (1972), have also confirmed the presence of antibodies against *R. slovaca* by serological analysis in small mammals from Slovakia.

Until now, mostly experimental studies of rickettsial infection of *A. flavicollis*, *Clethrionomys glareolus*, *M. arvalis* and *Mus musculus*, susceptibility and antibody formation were conducted in Slovakia (Řehácek et al., 1976, 1992).

In order to update information on the occurrence and diversity of rickettsiae, in the present study we focused on investigation of SFG *Rickettsia* spp. in the wild small mammals from natural, suburban and rural environments in eastern part of the country.

2. Material and methods

2.1. Sampling sites

Rodents and rodent-attached ticks were collected at three study sites in Eastern Slovakia during the years 2014–2016. The first study site, a Botanical garden with deciduous forest vegetation is located in the city of Košice (208 m a.s.l.; 48°446·84″N; 21°14′16·14″ E).

It represents an urban habitat with a significant anthropogenic impact. The second site, Čermeľ, is characterized by the presence of mixed forest vegetation, with a predominance of

beech, hornbeam and spruce (208–600 m a.s.l.; 48°45′46·67″N; 21°8′8·17″E), and it represents a suburban habitat. The third site, with mixed forest vegetation and a predominance of beech, oak and hornbeam, is located in the Slovak Karst National Park, near the Hrhov village (200–220 m a.s.l., 48°34.899 N, 20°46.743 E), and it represents a natural habitat.

2.2. Rodents sampling

In total, 100 Swedish bridge metal traps filled with sunflower seeds were set at each site, 5 m away from each other, for one night, and these were checked next morning. Captured small mammals were transported to the laboratory and euthanized in accordance with the licences of the Ministry of Environment of the Slovak Republic No. 4874/2011–2.2 and 4559/2015–2.3. Subsequently, they were examined for the presence of ectoparasites, which were removed, collected and stored in 70% ethanol.

Each individual was identified into species, weighed with an accuracy of ± 0.5 g, measured, euthanized and dissected and the sexual condition of each animal was ascertained. Classification of mammals into categories subadults (immature) and adults (mature) was used according to Pelikán (1965) with respect to the sexual condition of individuals. The removed ticks and other ectoparasites were determined with the use of an identification key (Filippova, 1977), by light microscopy. An ear biopsy was obtained from 245 animals and stored at −20 °C until further molecular analysis.

2.3. Molecular analysis

Genomic DNA was extracted from each ear-tissue sample after incubation with a Proteinase K solution at 56 °C overnight, using the NucleoSpin Tissue kit (Macherey-Nagel, Düren, Germany) according to the manufacturer's instructions.

The presence of *Rickettsia* spp. in the genomic DNA samples was evaluated by PCR (Polymerase Chain Reaction) amplification of the 381-bp long fragment of the citrate synthase gene (*gltA*), using the genus-specific primers RpCS.877p and RpCS.1258n (Regnery et al., 1991).

PCR amplifications were performed in a total of 25 µl of reaction mixture containing 14.9 µl of nuclease-free water, 2.5 µl of 10 × PCR buffer, 1 µl of 25 mM $MgCl_2$, 0.125 U of Taq DNA polymerase (Qiagen), 0.5 µl of 10 mM dNTP Mix, 0.5 µl of 10 µM primers and 5 µl of DNA template. Previously sequenced DNA of *Rickettsia* spp. isolated from infected rodents was used as a positive control and sterile water as a negative control in each reaction.

PCR products were analysed by electrophoresis in a 1.5% agarose gels stained with GoldView Nucleic Acid Stain (Beijing SBS Genetech, Beijing, China) and visualized with a UV transilluminator. After PCR analyses, several positive samples with amplified *gltA* gene fragments were purified for the sequencing. The aim was to choose the representatives from each location, and from different rodent species and sex. The PCR products selected for further phylogenetic analyses were purified using Isolate II PCR and Gel Kit (Bioline) and sequenced (at the University of Veterinary Medicine and Pharmacy in Košice, Slovakia) in both directions using the same primers as for the PCR amplification.

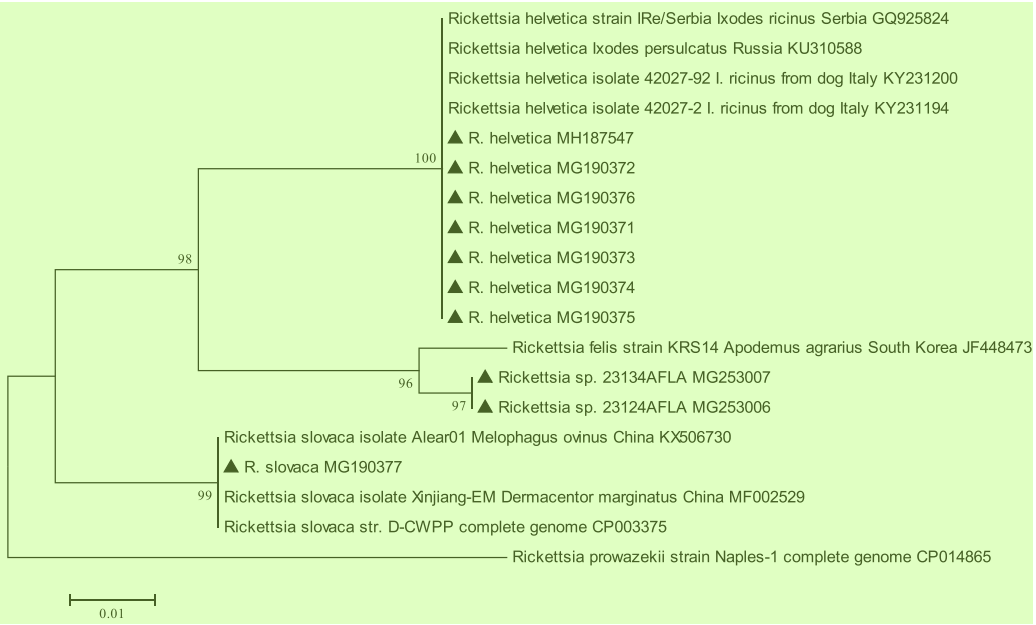


Fig. 1. Phylogenetic analysis of *Rickettsia* species.

The analysis involved 19 nucleotide sequences of *Rickettsia* spp. *gltA* gene fragment, ten obtained in this study and the representatives from GenBank database. The evolutionary history was inferred by using the Maximum Likelihood method based on the Tamura 3-parameter model (Tamura, 1992). The tree with the highest log likelihood is shown. The percentage of trees in which the associated taxa clustered together is shown next to the branches. Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Maximum Composite Likelihood (MCL) approach, and then selecting the topology with superior log likelihood value. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. The analysis involved 19 nucleotide sequences. *R. prowazekii* (CP014865) was used as an outgroup. All positions containing gaps and missing data were eliminated. There were a total of 271 positions in the final dataset. Evolutionary analyses were conducted in MEGA6 software (Tamura et al., 2013).

Nucleotide sequences obtained in this study are marked with black triangles

2.4. Phylogenetic analysis

The obtained nucleotide sequences of *Rickettsia* spp. *gltA* gene fragment were manually edited in MEGA 6 and compared with GenBank entries by BLAST (Basic Local Alignment Search Tool) (Tamura et al., 2013). Homologous sequences were aligned by the ClustalW program.

The evolutionary history of the sequences was inferred by using the Maximum Likelihood method based on the Tamura 3-parameter model (Tamura, 1992). The best DNA substitution model was chosen by Model Selection (ML) analysis. The tree with the highest log likelihood is shown (Fig. 1).

2.5. Statistical analysis

The associations between the prevalences of rickettsiae in males and females of the most abundant rodent species and the variation of seasonal rickettsial occurrence in rodents were evaluated by the Fisher's exact test using Past version 2.17b software (Hammer et al., 2001). A p value < 0.05 was considered to be statistically significant.

3. Results

A total of 250 rodents and shrews, belonging to eleven species (*Apodemus agrarius*, *Apodemus flavicollis*, *Apodemus uralensis*, *Myodes glareolus*, *Crocidura leucodon*, *Crocidura suaveolens*, *Micromys minutus*, *Microtus arvalis*, *Microtus subterraneus*, *Neomys fodiens* and *Sorex minutus*), were captured during the study. The majority of rodents was captured in the Čermeľ locality in a suburban environment (50.4%; n = 126), followed by the natural habitat near the village Hrhov in the Slovak Karst National Park (46.4%; n = 116). The lowest number of small mammals was captured in the urban landscape in the Botanical Garden within Košice (3.2%; n = 8). Two species, *A. agrarius* and *A. flavicollis*, prevailed at all sampling sites. *M. glareolus* was absent only in the rural habitat. Species composition and the number of captured individuals at the individual sampling sites are shown in Table 1.

In total, 1102 ixodid ticks were collected (920 larvae and 182 nymphs) from captured rodents. Based on morphological characteristics, at least five tick species belonging to genera *Ixodes*, *Dermacentor* and *Haemaphysalis* were identified.

The dominant tick species, *Dermacentor* sp. (52.1%, 427 larvae, 174 nymphs), was followed by *I. ricinus* (44.6%, 474 larvae, 18 nymphs), and *I. trianguliceps* (2.8%, 15 larvae, 16 nymphs). Moreover, four *H. concina* larvae and one *H. inermis* nymph were collected. In our investigation, larvae prevailed over nymphs. The highest diversity of tick species was recorded in the natural habitat near Hrhov village, followed by suburban area in Čermeľ, with two species, and the urban site, the botanical garden, with one recorded tick species. In total, 124 out of 250 captured hosts did not carry any tick.

Ear biopsies of 245 rodents, belonging to eight species, were analysed by PCR for the presence of rickettsiae. In total, 11% (27/245) of rodents tested positive for *Rickettsia* spp. In the Čermeľ locality the presence of *Rickettsia* spp. was confirmed in 16.0% (20/125) of rodents. In the natural habitat near Hrhov the prevalence of *Rickettsia* spp. reached 6.2% (7/113). In the urban area of the botanical garden, none of the seven examined species of rodents tested positive for rickettsiae.

We did not observe a significant difference in infection rates between males and females of small mammals ($p = 0.838$). Similarly, a sexual dimorphism was not confirmed in any of three dominant rodent species; *A. agrarius* ($p = 0.377$), *A. flavicollis* ($p = 0.320$), *M. glareolus* ($p = 0.352$) (Table 2).

Regarding the seasonal dynamics of rickettsial infection, the highest number of positive rodents was observed during the spring months, when the infection rate reached 15.3% and 21.1% in May and June, respectively; lower prevalence was recorded in September (6.1%) and the lowest prevalence rate was found in October (4.2%). By comparing the “spring-summer” months (May, June, July) with the autumn (September, October), a significant difference in infection rates of small mammals between seasons was confirmed ($p = 0.023$). On the contrary, seasonal differences were not confirmed for most abundant rodent species such as *A. agrarius* ($p = 1.00$), *A. flavicollis* ($p = 0.44$) and *M. glareolus* ($p = 1.00$). The detailed results on the seasonal dynamics of rickettsial infection in tested small mammals are presented in Table 3.

3.1. Phylogenetic analysis

Out of 27 *Rickettsia* spp. *gltA*-PCR- positive samples, 18 were chosen

Table 1
Small mammals captured in eastern Slovakia during.2014–2016

	Sampling season Location												
	2014				2015				2016				Total
	BG	Č	H	Σ	BG	Č	H	Σ	BG	Č	H	Σ	
Rodent species													
Apodemus agrarius	–	8	–	8	3	1	26	30	1	6	24	31	69
Apodemus flavicollis	–	47	–	47	1	–	14	15	2	8	27	37	99
Apodemus uralensis	–	–	–	–	–	–	–	–	–	–	1	1	1
Myodes glareolus	–	46	–	46	–	–	4	4	–	4	4	8	58
Crocidura leucodon	–	–	–	–	–	–	1	1	–	–	2	2	3
Crocidura suaveolens	–	–	–	–	–	–	–	–	1	–	–	1	1
Micromys minutus	–	–	–	–	–	–	–	–	–	4	–	4	4
Microtus arvalis	–	–	–	–	–	–	10	10	–	–	1	1	11
Microtus subterraneus	–	1	–	1	–	–	–	–	–	–	–	–	1
Neomys fodiens	–	1	–	1	–	–	–	–	–	–	–	–	1
Sorex minutus	–	–	–	–	–	–	–	–	–	–	2	2	2
Total	–	103	–	103	4	1	55	60	4	22	61	87	250

- rodent species not captured.
BG – Botanical garden; Č – Čermeľ; H - Hrhov.

Table 2
Presence of rickettsiae in tested small mammals captured in eastern Slovakia.

Species	No. of *Rickettsia*-positive (No. of sequenced) samples / No. of tested small mammals (males)	No. of *Rickettsia*-positive (no. of sequenced) samples / No. of tested small mammals (females)
Apodemus agrarius	0/43	1 (1)/26
Apodemus flavicollis	7 (4)/51	3 (3)/48
Apodemus uralensis	1 (1)/1	0
Myodes glareolus	5 (2)/30	8 (4)/28
Crocidura leucodon	0/2	1 (1)/1
Micromys minutus	0/3	0/1
Microtus arvalis	1 (1)/5	0/5
Microtus subterraneus	0	0/1
Total	14 (8)/135	13 (9)/110

for further sequencing. In total, 11 DNA amplicons were successfully sequenced. *Rickettsia helvetica* was the predominant species, with 7 out of 27 PCR- positive samples. It was confirmed in *A. agrarius* (1♀), *A. flavicollis* (1♀), *M. glareolus* (2♀, 2♂) and *M. arvalis* (1♀). *Rickettsia* sp., closely related to *R. felis*, was identified in *A. flavicollis* (2♀, 1♂) (3/27), and *R. slovaca* was found in only one *A. flavicollis* male (1/27). Sequencing of the remaining seven *gltA* positive samples failed, probably due to the low concentration of DNA.

BLAST (Basic Local Alignment Search Tool - NCBI) analysis of the *gltA* gene fragments confirmed the presence of *R. helvetica* in seven samples. Seven nucleotide sequences of *R. helvetica* were identical within the overlapping region and were submitted to GenBank. BLAST confirmed their 100% identity with at least 50 *R. helvetica* isolates from distinct vectors and/or hosts from all over the Europe (e.g. KY231194-KY231200; GQ925824; KJ40389, and others).

The nucleotide sequences of the *gltA* gene of *Rickettsia* sp. (23664AFLA, 23124AFLA, 23134AFLA) obtained in this study were 100% identical with isolate from the hedgehog flea (*Archaeopsylla erinacei*) removed from red fox in southern Slovakia (Víchová, unpublished data). Additionally, these sequences showed 99% identity with *R. felis* strain KRS14 (JF448473) isolated from *A. agrarius* in South Korea.

Similar results were published by Špitalská et al. (2015). Phylogenetic analysis based on the fragment of *gltA* gene nucleotide sequences showed that *Rickettsia* sp. detected in *C. solutus* flea collected from *A. agrarius* in Eastern Slovakia clustered together with the same *R. felis* Korean isolate. Unfortunately, the authors did not insert the mentioned sequences in the GenBank database; thus, it was impossible to compare them with our *Rickettsia* sp. sequences.

The nucleotide sequence of the *gltA* gene of *R. slovaca* obtained in this study was 98% identical with the *R. slovaca* isolate from the sheep ked (*Melophagus ovinus*) from China (KX506730). Altogether, 18 nucleotide sequences of *Rickettsia* spp. *gltA* gene fragment with a length of 271 bp, including sequences obtained in this study as well as those retrieved from the GenBank, were used for a subsequent phylogenetic

Table 3
Seasonal occurrence of *Rickettsia*-infected rodents in eastern Slovakia during 2014–2016.

Rodent species	Season Prevalence (%) of *Rickettsia* spp. (No. of *Rickettsia*-positive rodents/No. of tested rodents)					
	May	June	July	September	October	Total
Apodemus agrarius	0 (0/9)	0/1	–	2.1 (1/48)	0 (0/11)	(1/69)
Apodemus flavicollis	9.4 (5/53)	25 (4/16)	0/5	5.5 (1/18)	0 (0/7)	(10/99)
Apodemus uralensis	–	–	–	0 (1/1)	–	(1/1)
Myodes glareolus	24.5 (12/49)	0 (0/2)	0/3	25 (1/4)	–	(13/58)
Crocidura leucodon	–	–	–	33.3 (1/3)	–	(1/3)
Micromys minutus	–	–	–	0 (0/4)	–	(0/4)
Microtus arvalis	–	–	–	0 (0/4)	16.6 (1/6)	(1/10)
Microtus subterraneus	0 (0/1)	–	–	–	–	(0/1)
Total	15.3 (17/111)	21.1 (4/19)	(0/8)	6.1 (5/82)	4.2 (1/24)	11 (27/245)

- rodent species not captured.

analysis.

All *R. helvetica* sequences (MG190371-376, MH187547) obtained in the present study are placed in a single, well-supported cluster with other *R. helvetica* isolates from *Ixodes* spp. ticks (e.g. from Italy, Serbia and/or Russia) (Fig. 1).

The second clade consists of *R. slovaca gltA* nucleotide sequences and comprises isolate (MG190377) from the yellow-necked mouse from eastern Slovakia as well as sequences isolated from the sheep ked (*Melophagus ovinus*) (KX506730), from *Dermacentor marginatus* (MF002529) from China and/or the *R. slovaca* type strain D-CWPP (CP003375) (Fig. 1).

Two nucleotide sequences, designated here as *Rickettsia* sp. 23134AFLA (MG253007) and 23124AFLA (MG253006) from the yellow-necked mice, are closely related (98.5% identity) with the *R. felis* strain isolated from the striped field mouse (*A. agrarius*) in South Korea and are identical with the isolate from the hedgehog flea (*A. erinacei*) removed from red fox in Southern Slovakia (data not shown). These nucleotide sequences created a separate cluster in the phylogenetic tree (Fig. 1).

3.2. Nucleotide sequences accession numbers

Nucleotide sequences of the fragments of *gltA* genes obtained in this study were submitted to the GenBank database under the following accession numbers: *R. helvetica* MG190371, MG190372, MG190373, MG190374, MG190375, MG190376, MH187547, *R. slovaca* MG190377, *Rickettsia* sp. 23124AFLA (MG253006), and *Rickettsia* sp. 23134AFLA (MG253007).

4. Discussion

Small mammals are associated with several pathogenic microorganisms and host a wide spectrum of ectoparasites. Some species of rodents (e.g. *A. flavicollis*) in particular are more heavily infested with *Ixodes* spp. ticks than others (Stanko and Miklisová, 2000). In general, a different composition of rodent species was found in individual habitats, suggesting that the character of the environment and the structure of habitat considerably influence the composition of the hosts and their ectoparasites. In total, 250 individuals of 11 species were captured, with *A. agrarius* and *A. flavicollis* being the most prevalent species at all three sampling sites during a three-year study.

Similarly, as it has been observed in the present study, the two above-mentioned *Apodemus* species prevailed in the lowland habitats of eastern Slovakia even in the past (Stanko and Miklisová, 2000). *M. glareolus* was absent only in the urban environment, in the botanical garden. The highest diversity of murid species as well as the highest diversity of tick species feeding on rodents was observed in the natural habitat, in the Slovak Karst National Park region. In addition, *M. arvalis* occurred solely at this natural site, preferring pastures with shrubs.

In this study, the wild small mammals were screened for the presence of *Rickettsia* spp. in three different types of habitat in eastern Slovakia. The highest prevalence of *Rickettsia* spp. was observed in rodents captured in the suburban habitat. Rickettsial DNA was detected in ear samples of small mammals belonging to six species, namely *A. agrarius*, *A. flavicollis*, *A. uralensis*, *M. glareolus*, *M. arvalis*, and *C. leucodon*. *Rickettsia* sp. closely related to *R. felis* and two pathogenic species, *R. helvetica* and *R. slovaca*, were identified in *A. agrarius*, *A. flavicollis*, *M. glareolus* and *M. arvalis*.

The highest prevalence of rickettsiae was recorded in *M. glareolus* (13/58), followed by *A. flavicollis* (10/99). Only one rodent species, specifically *A. flavicollis* captured in the suburban and rural areas, carried DNA of different *Rickettsia* species (*R. helvetica*, *R. slovaca*, and *Rickettsia* sp. closely related to *R. felis*), suggesting that some species of rodents might play a more important role as hosts for different rickettsiae. The data indicate that bank voles and yellow-necked mice are probably more often infected with rickettsiae than other species of rodents.

Rickettsiae are considered to be emerging or re-emerging pathogens, and there are limited data on their presence and prevalence in many regions of Europe (Sprong et al., 2009). Up to now, several cases of human rickettsioses have been recorded in Hungary (Lakos, 1997), France (Raoult and Roux, 1997; Raoult et al., 2002), Sweden (Nilsson et al., 1999, 2010), Switzerland (Baumann et al., 2003), Bulgaria (Komitova et al., 2003), Portugal (de Sousa et al., 2006, 2008), Italy (Selmi et al., 2008), Spain (Aguirrebengoa et al., 2008), Poland (Świtaj et al., 2012) and Slovakia (Sekeyová et al., 2012).

In the present study, the overall prevalence of *Rickettsia* spp. in wild small mammals reached 11%, without a significant difference in infection rates between males and females ($p = 0.838$).

The highest occurrence of *Rickettsia*-infected rodents in this study was observed during the spring in May–June (15.3–21.1%) and during the fall in September (6.1%), followed by a decreasing number of infected rodents to a minimum of 4.2% in October (Table 3). Similarly, the study of Špitalská et al. (2016) mentioned that the seasonal dynamics of *Rickettsia*-infected

I. ricinus in Slovakia showed two major peaks, in spring (April–June/6.8%–9.2%) and autumn (October 7.8%). The seasonal dynamics and the rates of *Rickettsia*-infection in small mammals is related to the activity of their tick vectors.

Among the successfully genotyped samples, *R. helvetica* was the most prevalent species recorded in small mammals at two sampling sites, followed by *Rickettsia* sp. closely related to *R. fellis*. *R. slovaca* was confirmed in one of successfully sequenced samples.

In study from Poland (Biernat et al., 2016) none of the examined *A. flavicollis* and *M. glareolus* blood samples tested were positive for *Rickettsia* spp., but rickettsial DNA was identified in rodent-attached *I. ricinus* ticks. The authors suggested a very short-lasting rickettsiemia in small mammals or the possibility of vertical transmission of infection.

The presence of *R. slovaca* was originally confirmed in vector ticks *D. marginatus* and *D. reticulatus* (Brezina et al., 1969; Špitalská et al., 2012; Minichová et al., 2017). As mentioned in the introduction, *R. slovaca* was identified in ticks and small mammals by serological or microscopic methods in the past, but our study represents the first molecular evidence of *R. slovaca* in wild small mammals in Slovakia. In our study, *R. slovaca* was detected in *A. flavicollis* captured in the Slovak Karst region, where the sympatric occurrence of *D. marginatus* and *D. reticulatus* ticks, the main vectors of *R. slovaca*, is quite common. A previous examination of a smaller sample of *D. reticulatus* ticks from the same area, Slovak Karst (Hrhov village) region, confirmed 8.1% (6/74) prevalence of rickettsiae in adult ticks. Among other species of rickettsiae, the presence of *R. slovaca* was recorded too (Radzijevskaja et al., 2015). We assume a higher occurrence of *R. slovaca* in small mammals (and other potential vertebrate hosts), especially in areas with the presence of *D. marginatus* and *D. reticulatus* ticks. Species composition of vectors and their hosts influence the occurrence and distribution of pathogens in the studied area.

To analyze the tick species composition and the dominance of individual tick species on small mammals in detail was not the aim of this study. However, the presence of several tick species, e.g. *Ixodes* spp., *Dermacentor* spp. and *Haemaphysalis* spp. on small mammals in the Slovak Karst region, reflects the sympatric occurrence of distinct tick species at the sampling site. This fact has already been published in the past (e.g., Nosek, 1971, 1972; Hubálek, 1986, and others).

The investigation of small-mammal populations for the presence of parasites and pathogens can help to improve knowledge about the occurrence, circulation and spreading of infectious agents in a specific area. The data obtained in this study suggest that small mammals, mainly some predominant species (e.g. *Apodemus* spp., *Myodes* spp.), might play a more important role in the maintenance, circulation and dissemination of rickettsiae within and outside the natural foci. The prevalence of rickettsiae in small mammals depends on the type of habitat, species composition and seasonal activity of vectors and

vertebrate hosts, as well as on the fluctuations of small-mammal populations.

The herein presented results suggest a higher risk of infection with zoonotic pathogens, *R. helvetica* and *R. slovaca* in natural and suburban habitats of Eastern Slovakia, especially during the spring and autumn season.

5. Conclusion

The presence of *R. helvetica*, *R. slovaca* and *Rickettsia* sp. closely related to *R. felis* was confirmed in small mammals in Eastern Slovakia. *R. helvetica* was found to be the predominantly occuring species in the two investigated sites from several host species. *R. slovaca* was detected in *A. flavicollis* captured in the Slovak Karst region, where the sympatric occurrence of *D. marginatus* and *D. reticulatus* ticks is confirmed. We assume the presence of *R. slovaca* especially in sites with the occurrence of *Dermacentor* spp. In this study, *Rickettsia* sp. closely related to *R. felis* was first time detected in *A. flavicollis* in suburban site of Slovakia. Some rodent species that have a wider ecological valency may contribute to the maintenance, circulation and dissemination of rickettsiae within and out the natural foci more significantly than those species that have narrower relation to the certain type of habitat.

The yellow-necked mouse (*A. flavicollis*) might be an important reservoir host of different *Rickettsia* spp., not only in Slovakia.

Conflict of interest statement

The authors declare no conflict of interest.

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